



Subject: Maths (MAT), Course Number: 142
Assignment 1 (20 points)

Due on Friday, September 8th at 11:59 pm

Answer all questions. Choose the most appropriate option for each multiple-choice question.

Q1. The roots of the quadratic equation $x^2 - 5x + 6 = 0$ are:

- A. (1, 6)
- B. (2, 3)
- C. (3, 5)
- D. (-2, -3)

Q2. The simple interest on Rs. 5000 at 10% per annum for 2 years is:

- A. Rs. 1000
- B. Rs. 1050
- C. Rs. 1100
- D. Rs. 1200

Q3. The derivative of $\ln(x^2 + 1)$ with respect to x is:

- A. $\frac{1}{x}$
- B. $\frac{2x}{x^2 + 1}$
- C. $\frac{1}{x^2 + 1}$
- D. $2x$

Q4. The exact value of $\cos(2\theta)$ is:

- A. $1 - 2\sin^2 \theta$
- B. $\sin^2 \theta + \cos^2 \theta$
- C. $2\sin \theta \cos \theta$
- D. $\tan \theta$

Q5. A fair die is rolled. The probability of getting a number greater than 4 is:

- A. $\frac{1}{6}$
- B. $\frac{2}{6}$
- C. $\frac{3}{6}$
- D. $\frac{4}{6}$

Q6. The determinant of the identity matrix of order 3 is:

- A. 0
- B. 1
- C. 2
- D. 3

Q7. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} =$

- A. 0
- B. $\frac{1}{2}$
- C. 1
- D. ∞

Q8. If a sum of Rs.10,000 amounts to Rs.12,100 in 2 years at compound interest, the rate per annum is:

- A. 5%
- B. 10%
- C. 12%
- D. 15%

Q9. $\sin^2 \theta + \cos^2 \theta =$

- A. 0
- B. 1
- C. $\sin 2\theta$
- D. $\cos 2\theta$

Q10. The sum of the first n terms of an arithmetic progression with first term a and common difference d is:

- A. $\frac{n}{2} [2a + (n - 1)d]$
- B. a^n
- C. $n(a + d)$
- D. $(a + d)^n$